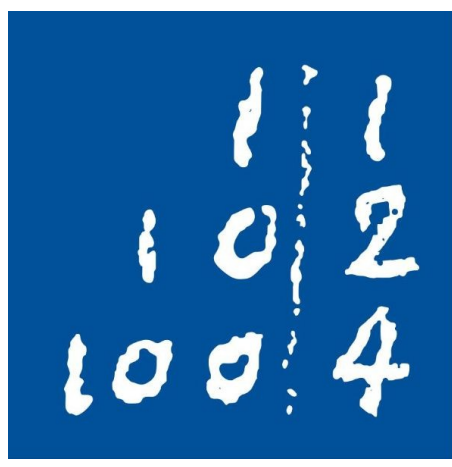


Memory Reach: GRU vs MLP under PPO on MiniGrid Memory

Anh Khoa Pham



Leibniz
Universität
Hannover

Code, studies and
winning checkpoints



1 Motivation

- **POMDP**: only a 7×7 view is seen
- **Memory** must live in a state h
- **Full BPTT** costs memory $\propto$ length
- **Truncation** cuts gradients to L
- L is inherited, never measured

Core question: how far back must the gradient reach, and what does that reach cost?

250 runs · $\approx$ 236 GPU-hours · RTX 3060 12 GB

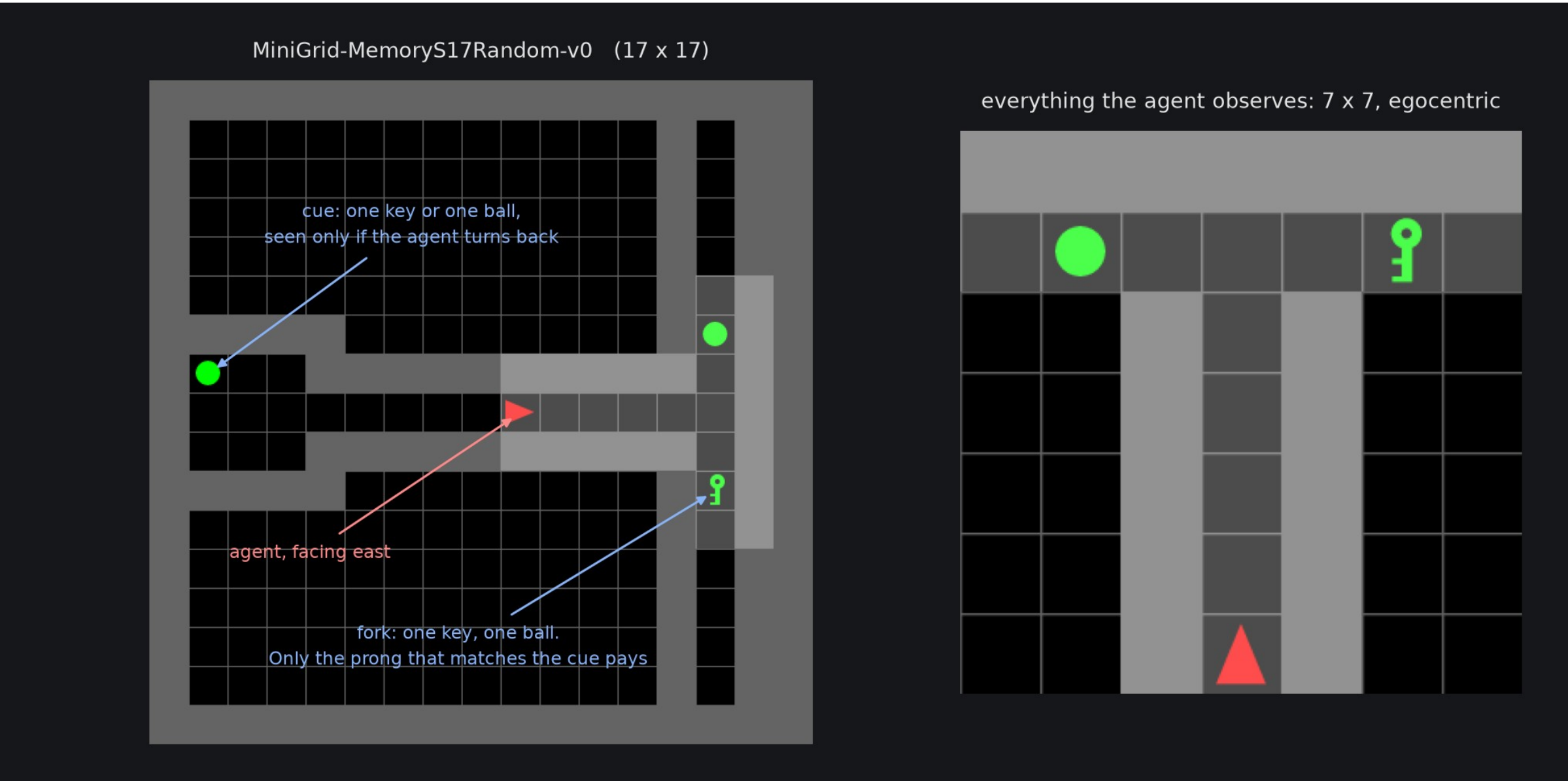
2 Problem Setting

The environment

- MiniGrid-MemoryS17Random-v0
- Cue (key or ball) in the west room
- T-junction at the far east end
- Only the matching prong pays
- Agent spawns facing away from it
- Both prongs look identical there
- Reward $1 - 0.9 \cdot \text{steps} / \text{max_steps}$

The two encoders

- **MLP**: no state, gradient spans 1 step
- **GRU**: carries h , gradient reaches L



Left: the maze. Right: everything the agent sees.

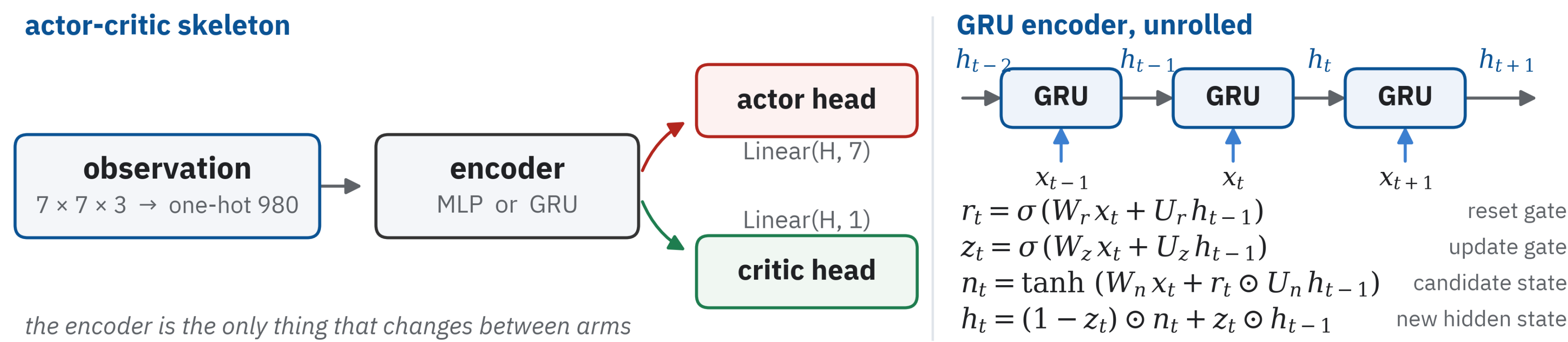
6 Limitations & Future Work

- One task, one architecture, one algorithm
- $L \in \{5, 6, 7\}$ never run: threshold bracketed
- Next: sweep L and fix the corridor length
- Then: R2D2 burn-in, LSTM, small transformer
- Five seeds, one evaluation seed of 50 mazes

References. Hausknecht & Stone 2015 · Kapturowski et al. 2019 · Ni et al. 2022 · Schulman et al. 2017 · Chevalier-Boisvert et al. 2023

3 Approach

Network



PPO objective

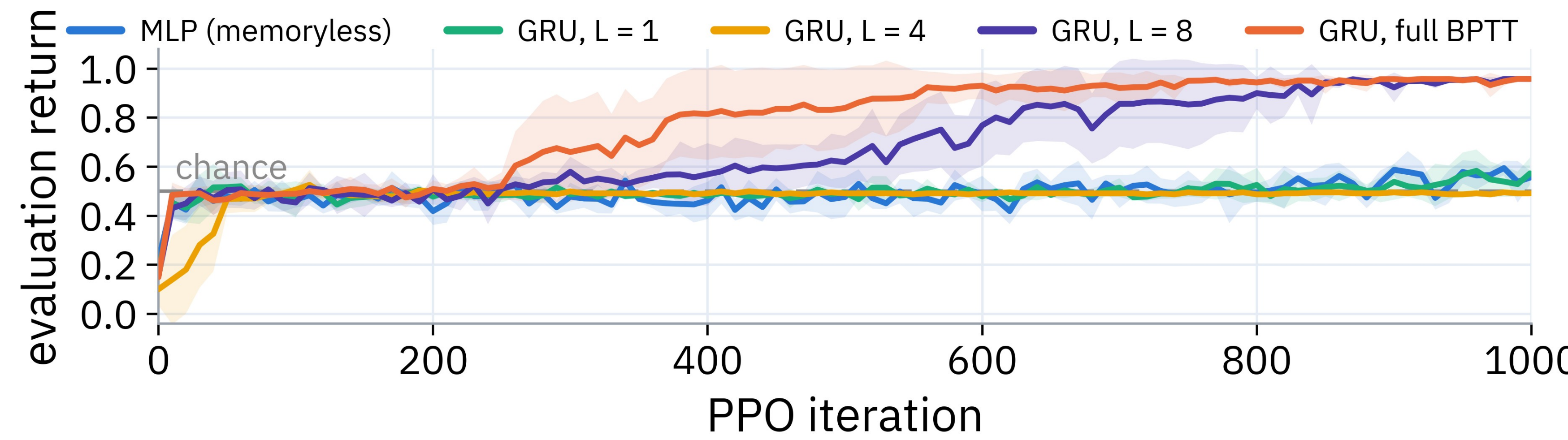
$$\begin{aligned} \rho_t &= \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t) && \text{probability ratio} \\ L^\pi &= -E_t[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \hat{A}_t)] && \text{clipped policy loss} \\ L^V &= E_t[(V_\theta(s_t) - \hat{R}_t)^2] && \text{value loss} \\ L &= L^\pi + c_v L^V - c_e H[\pi_\theta] && \text{what is backpropagated} \end{aligned}$$

Hyperparameter Optimization

- **Surrogate model:** Optuna TPE sampler, seeded at 42
- **Budget:** 50 trials per arm $\times$ 5 arms = 250 training runs
- **Seeds per trial:** 5, each 1000 iterations $\approx$ 11.6M steps
- **Objective:** mean – std of return over the 5 seeds

4 Results

arm	return	success rate	steps	minibatch	time
MLP (memoryless)	0.556 ± 0.029	0.592 ± 0.027	24.3 ± 2.3	1024	1.3 h
GRU, $L = 1$	0.573 ± 0.064	0.588 ± 0.068	9.8 ± 4.1	4096	2.6 h
GRU, $L = 4$	0.492 ± 0.001	0.500 ± 0.000	7.0 ± 0.0	4096	1.6 h
GRU, $L = 8$	0.958 ± 0.004	1.000 ± 0.000	16.8 ± 1.4	4096	2.0 h
GRU, full BPTT	0.958 ± 0.002	1.000 ± 0.000	16.8 ± 0.9	512	1.3 h



Winning trial per study; mean $\pm$ std across the five seeds.

5 Key Insights

- 1 **Carrying h forward is not what matters.**
Every GRU arm does it; only the reach of the gradient differs.
- 2 **What must fit inside L is the gap, not the episode.**
A gap ≤ 8 is reachable in 62 % of mazes, a gap ≤ 4 in only 26 %.
- 3 **The failing arms do not remember partially.**
 $L = 4$ always picks the same prong: exactly 25/50, std 0.000.
- 4 **The reach costs memory, not time.**
Full BPTT is the fastest GRU arm (1.3 h); it pays in VRAM.

Recommendation. Full BPTT; $L = 8$ on a smaller GPU.